

Course Introduction

Parallel and Heterogeneous Computing

David Mayerich

Scalable Tissue Imaging and Modeling (STIM) Laboratory

Department of Electrical and Computer Engineering

Cullen College of Engineering

University of Houston

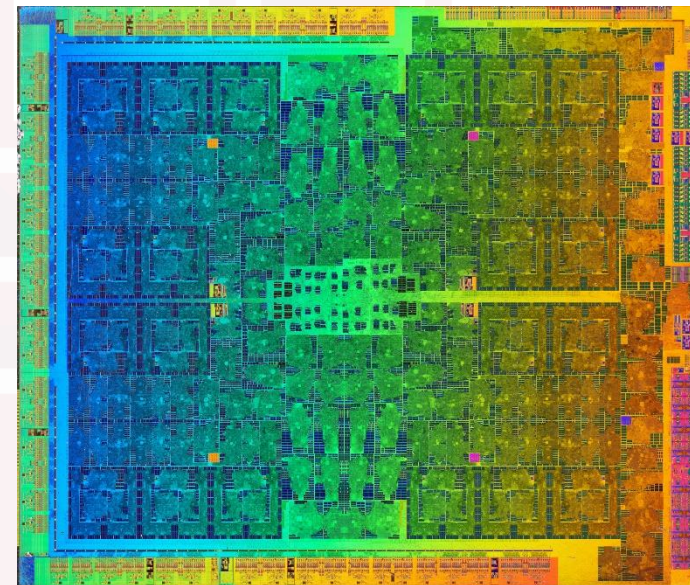
UNIVERSITY of **HOUSTON** | ECE

Goals and Expectations

- Parallel hardware from the POV of a developer
- Principles of parallel programming
- Multi-core CPU programming with C++
- Data-parallel GPU programming with CUDA
- Designing parallel algorithms
- Profiling CPU and GPU code to maximize throughput
- Popular libraries for parallel software development

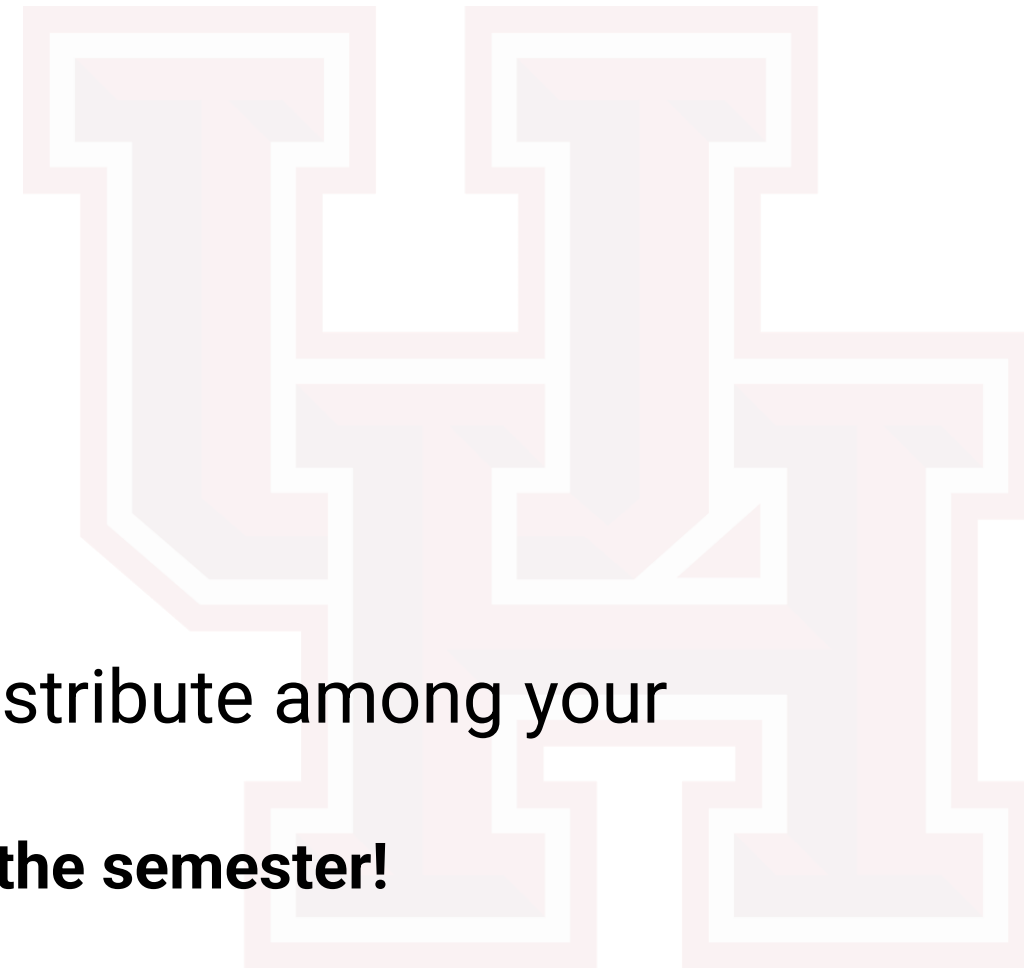


CuPy



Grading

- 5% Written Homework (25% undergrad)
- 25% Mid-Term Exam
- 25% Programming Assignments
- 25% Final Exam
- 20% Final Project (graduate only)
- You have 8 “late days” that you can freely distribute among your programming and project assignments
 - **Please note I can't delay these past the end of the semester!**

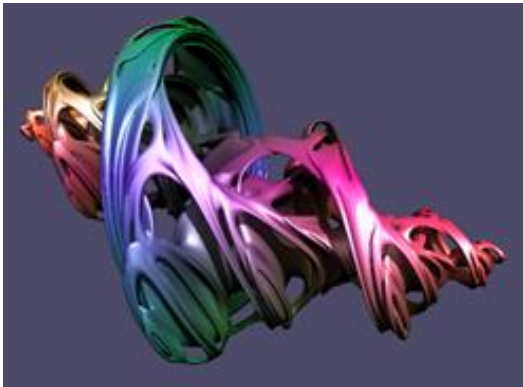


Homework Assignments

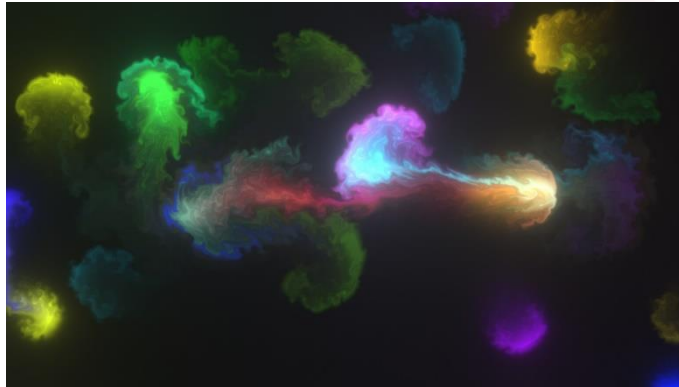
- 5% (25% for undergraduates) – primarily practice for exams
- Material from class lectures and reading
- Submission: Canvas, submit as a PDF
- Please ask (and answer) homework-related questions on Teams

Programming Assignments

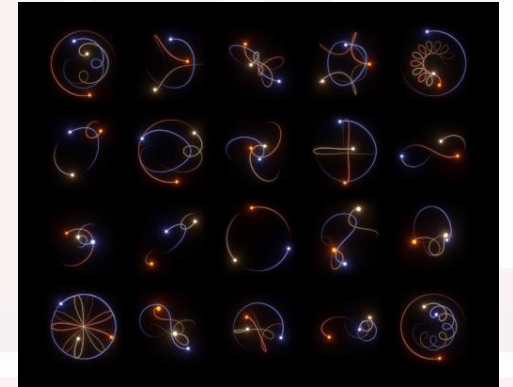
- Programming must be done on your own
- We will pick a programming assignment and develop it incrementally:
 1. Algorithm development and CPU optimization
 2. Multi-threading and CPU-level parallelism
 3. SIMD parallelism and GPU programming



ray tracing



Navier-Stokes simulation



n-body simulation

Exams

- Exams are comprehensive
 - 25% mid-term exam (primary focus: parallel concepts, threads)
 - 25% final exam (primary focus: CUDA)
- Code in exams:
 - Significant portions of this class focus on programming and algorithms
 - I will ask you to write code on exams BUT I don't expect perfection
 - Concepts are important – function names, signatures, parameters are secondary

Programming Assignment Submissions

- Submit your source repository as a zip file on Canvas
 - The unzipped folder should work as a source directory in CMake
- Verify your code on the following platform (at least):
 - GNU GCC 13
 - Stick to the C++20 standard
 - CMake 3.28
 - CUDA Toolkit 12 (note: macOS doesn't support CUDA)
 - (These are the defaults for Ubuntu 24.04 LTS)
- I may run your code on GNU in class to keep you on your toes!
- Don't post code publicly (ex. GitHub) until the course is over

Campus Resources

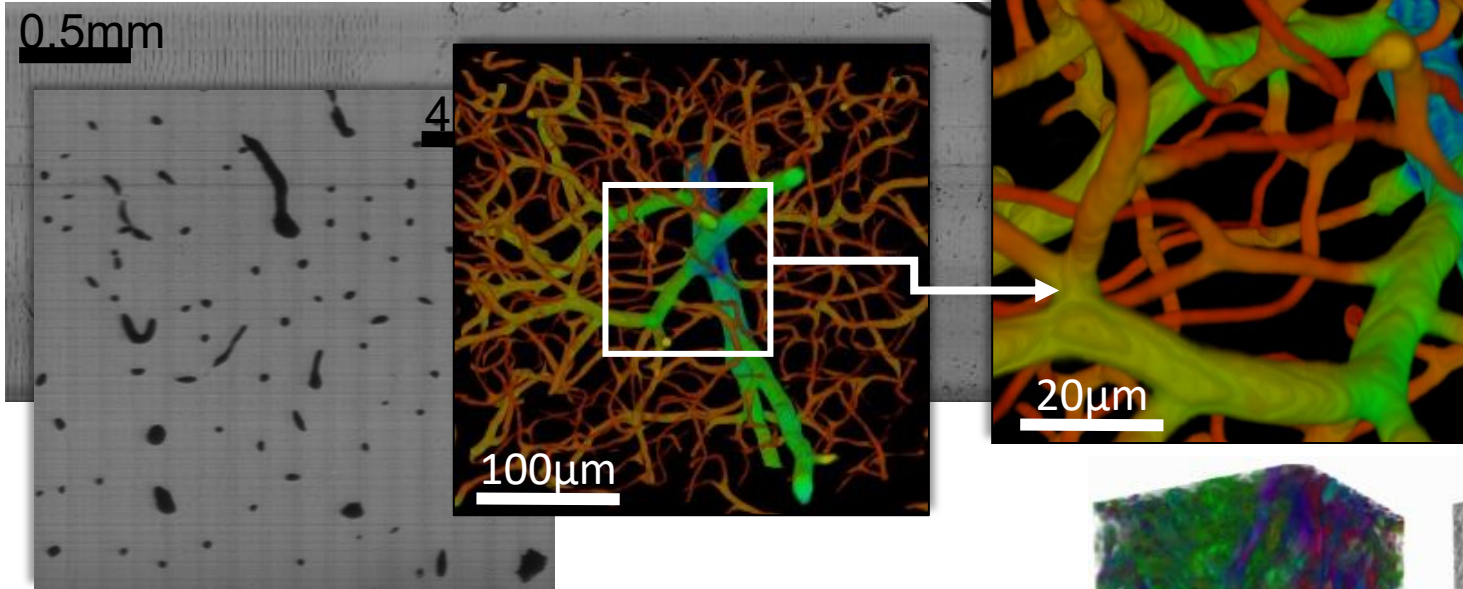
- UH Research Computing Data Core
 - Sabine Cluster: NVIDIA P100s
 - Carya Cluster: NVIDIA V100s
- STIM Lab: NVIDIA Geforce 3090
- HPE-DSI 101
 - Introduction to Cluster Computing
 - Linux, scripting, batch processing basics
- NSF Access: I can add students to an allocation assigned to this course
- The Cherno (YouTube)
 - [C++ Tutorial Playlist](#)
 - [OpenGL Tutorial Playlist](#)

Prerequisites

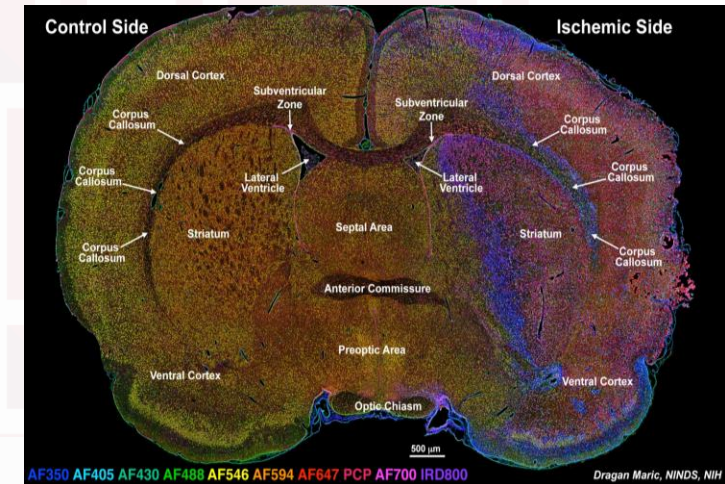
- ECE 3331 Programming Applications
 - Programming in C or C++
 - Memory management and pointers
 - Using a compiler and linker
- MATH 3321 Engineering Math
 - Differential and integral calculus
 - Solving differential equations
 - Linear algebra and Hilbert Spaces
 - Fourier Transforms
- ECE 3340 Numerical Methods
 - Floating point numbers
 - Taylor series
 - Asymptotic analysis (ex. big- O)
 - Loss of precision
- Hardware and software
 - Multi-Core CPU
 - CUDA (Nvidia) GPU
 - Understand your development tools
 - Understand your package manager

Scalable Tissue Imaging and Modeling (STIM) Lab

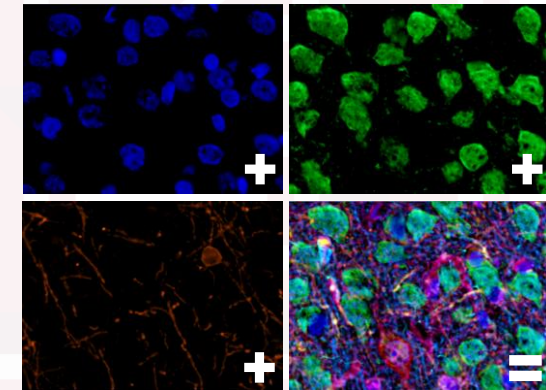
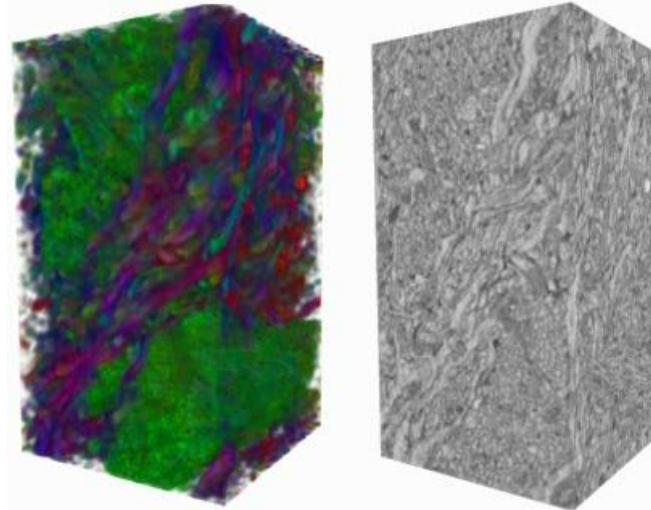
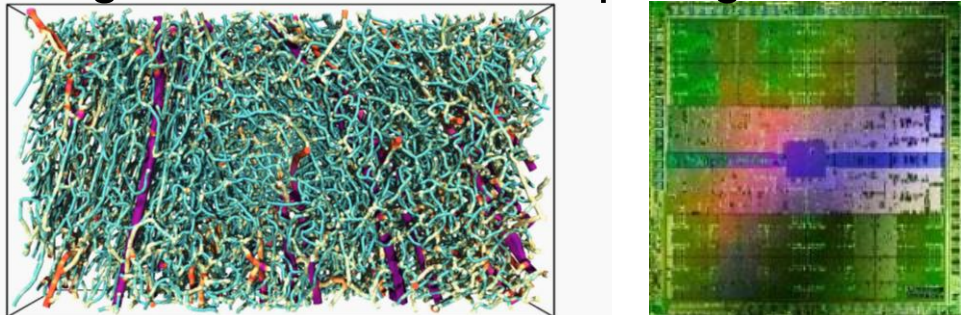
Large Scale Microscopy



Immunohistochemistry



High Performance Computing



About Me

- BS Computer Science – SWOSU (Oklahoma)
- PhD Computer Science – Texas A&M
- Beckman Fellowship – University of Illinois
- Background: computer graphics, microscopy
- Personal
 - Video games, TTRPG
 - Woodworking
 - Fencing

